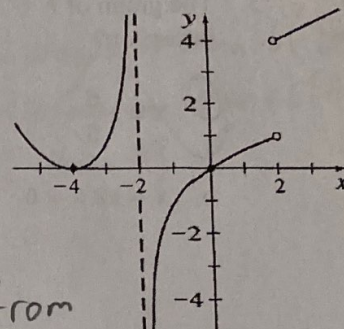


Name: [REDACTED]

Date: [REDACTED]

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1. Suppose f is a function that is continuous for all real numbers. The graph of its derivative function $f'(x)$ is given.



- a) List all critical values of f .

$x = -4, -2, 0, 2$

- b) At what value(s) of x , if any, does the graph of $f(x)$ have a local minimum? Justify your response.

When $x=0$ is a minimum because $f'(x)$ goes from negative to positive, meaning $f(x)$ goes from decreasing to increasing, therefore making it a local minimum.

- c) On what intervals is the graph of $f(x)$ Concave Up? when $f'(x)$ is increasing

when $x \in (-4, -2), (-2, 2), (2, \infty)$

2. Explain the similarities and differences between using "The First Derivative Test" and "The Second Derivative Test". Use proper terminology and be specific.

Similarities

Both require finding the first derivative and the critical values of the first derivative. Both tests are done by finding positive/negative and relating it back to the original function. Both can be used to find local max and min. (SDT $\rightarrow$ 2nd derivative test, FDT $\rightarrow$ 1st derivative test)

Differences

Second derivative is used in SDT, but not in FDT. FDT finds increasing/decreasing intervals while SDT finds concavity at critical points. SDT may fail (by getting 0/DNE) where FDT doesn't.

Multiple Choice: Choose the best answer for the following questions:

3. If $f'(x) > 0$ for all x , then

a. $f(x_1) = f(x_2)$ for every $x_1 < x_2$

c. $f(x_1) > f(x_2)$ for every $x_1 < x_2$

b. $f(x_1) < f(x_2)$ for every $x_1 > x_2$

d. $f(x_1) > f(x_2)$ for every $x_1 > x_2$

4. What is the behaviour of $f'(x)$ if $f''(x) > 0$?

- a. Increasing
b. Concave up
c. Concave down
d. Need more information

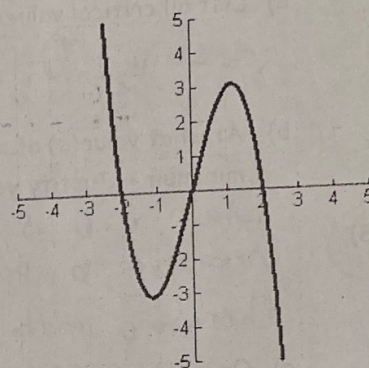
5. The graph of $f''(x)$ is displayed to the right. For what value(s) of x does $f(x)$ have a point of inflection?

a. $x = \pm \frac{6}{5}$

c. $x = \pm 2$

b. $x = \pm 2, x = 0$

d. $x = 0$



6. Given, $f(x) = \frac{x^3}{1-x^2}$, sketch the graph of $f(x)$ by

considering:

a) First derivative; intervals of increase/decrease and local extrema

$$f'(x) = \frac{3x^2(1-x^2) + 2x(x^3)}{(1-x^2)^2} = \frac{-x^4 + 3x^2}{(1-x^2)^2}$$

(3)

$$= \frac{-x^2(x^2-3)}{(1-x^2)^2} \rightarrow x = 0, \pm\sqrt{3}$$

$$\rightarrow x = \pm 1 \text{ (VA)}$$

$$f(-\sqrt{3}) = \frac{(-\sqrt{3})^3}{1-3} = \frac{-5.2}{-2} = 2.6$$

$$f(\sqrt{3}) = -2.6$$

local max: $(\sqrt{3}, -2.6)$

local min: $(-\sqrt{3}, 2.6)$

$f(x)$ increasing when $x \in (-\sqrt{3}, -1), (-1, 0), (0, 1), (1, \sqrt{3})$

$f(x)$ decreasing when $x \in (-\infty, -\sqrt{3}), (\sqrt{3}, \infty)$

	$(-\infty, -\sqrt{3})$	$(-\sqrt{3}, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \sqrt{3})$
f'	-	+	+	+	+
f	↓	↑	↑	↑	↑
		min $(-\sqrt{3}, 2.6)$	VA	HT	VA
				max $(\sqrt{3}, -2.6)$	

extended
 $(\sqrt{3}, \infty)$

$$f'(x) = \frac{3x^2 - x^4}{(1-x^2)^2} \quad f(x) = \frac{x^3}{1-x^2}$$

b) Second Derivative; intervals of concavity and point(s) of inflection

$$f''(x) = \frac{(6x - 4x^3)(1-x^2)^2 - 2(1-x^2)(-2x)(3x^2 - x^4)}{(1-x^2)^4}$$

$$= \frac{-2(1-x^2)((2x^3 - 3x)(1-x^2) - 2x(3x^2 - x^4))}{(1-x^2)^4}$$

$$= \frac{-2(2x^3 - 2x^5 - 3x + 3x^3 - 6x^3 + 2x^5)}{(1-x^2)^3}$$

$$= \frac{-2(-4x^3 - 3x)}{(1-x^2)^3} = \frac{2x(x^2 + 3)}{(1-x^2)^3} \rightarrow x=0 \quad x=\pm 1$$

concavity			
f''	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$
	+	-	+
f	cu	cd	cu
	VA	POI	VA
		$(0, 0)$	

$f(x)$ concave up when $x \in (-\infty, -1), (0, 1)$

$f(x)$ concave down when $x \in (-1, 0), (1, \infty)$

POI: $(0, 0)$

c) Y-intercept, asymptotes and behaviours around the asymptotes

$$f(0) = 0 \quad y\text{-int} = (0, 0)$$

$$VA: x = \pm 1$$

$$\lim_{x \rightarrow -1^-} f(x) = \infty \quad \lim_{x \rightarrow -1^+} f(x) = -\infty$$

$$\lim_{x \rightarrow 1^-} f(x) = \infty \quad \lim_{x \rightarrow 1^+} f(x) = -\infty$$

$$\frac{-x}{\sqrt{x^3 + 0x^2 + 0x + 0} - x^3 + 0x^2 - x} \rightarrow \text{call this } h(x)$$

oblique = 0 when $x = 0$

$x \rightarrow$ cross at $(0, 0)$

$$h(100) = -100 \quad f(100) = -100.01$$

$$h(-100) = 100 \quad f(-100) = 100.01$$

$$\lim_{x \rightarrow \infty} f(x) = -\infty \quad \lim_{x \rightarrow -\infty} f(x) = \infty$$

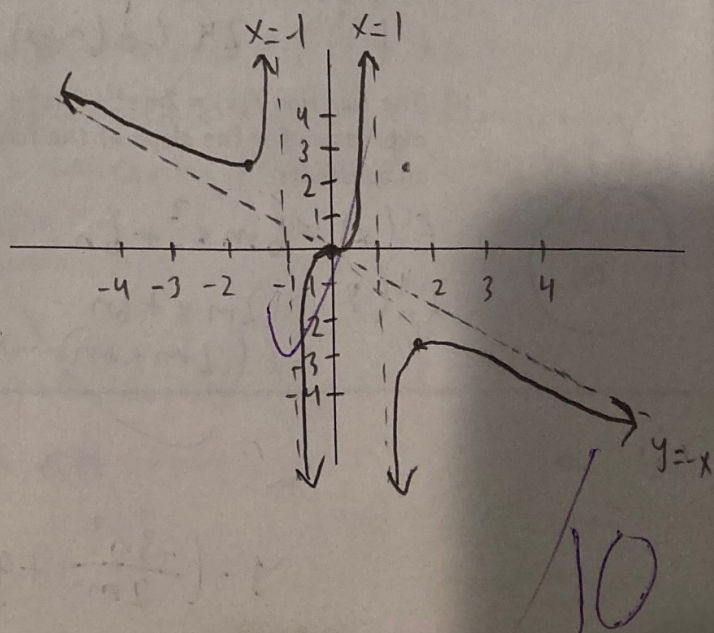
below

above

(2) Graph

$$\max: (\sqrt{3}, -2.6)$$

$$\min: (-\sqrt{3}, 2.6)$$



7. Find $f(2)$ if $(4, -7)$ is a point of inflection for $f(x) = x^3 + cx^2 + x + d$?

$$\begin{aligned} f'(x) &= 3x^2 + 2cx + 1 & -7 &= 64 + 192 + 4 + d \\ f''(x) &= 6x + 2c & 117 &= d \\ 0 &= 24 + 2c & f(x) &= x^3 - 12x^2 + x + 117 \\ -24 &= 2c & f(2) &= 8 - 48 + 2 + 117 \\ c &= -12 & \boxed{f(2) &= 79} \end{aligned}$$

8. Determine whether or not the function $f(x) = \sqrt{x^2 + 1}$ lies above or below the tangent at x

$x = 4$

$$f'(x) = \frac{1}{2} \frac{2x}{\sqrt{x^2 + 1}} = \frac{x}{\sqrt{x^2 + 1}} \quad f'(4) = \frac{4}{5} \quad f(4) = \sqrt{17} = 4.12$$

$$y = \frac{4}{5}x + b \quad b = -3.03 \quad y = \frac{4}{5}x - 3.03$$

$$\sqrt{17} = \frac{16}{5} + b \quad h(4) = \frac{16}{5} - 3.03 = 4.13$$

$\therefore$ below b/c $f(4) < h(4)$

9. Classify the local extrema as minimum or maximum for the graph of $f(x) = -x^3 + 9x^2 + 21x - 27$ without using the "first derivative test".

$$\begin{aligned} f'(x) &= -3x^2 + 18x + 21 \\ &= -3(x^2 - 6x - 7) \\ &= -3(x-7)(x+1) \rightarrow x=7, x=-1 \end{aligned}$$

$$\begin{aligned} f(-1) &= 1 + 9 - 21 - 27 = -38 \\ f(7) &= 218 \end{aligned}$$

$$\begin{aligned} f''(x) &= -6x + 18 \\ f''(-1) &= 24 \text{ (u min)} \\ f''(7) &= -24 \text{ (d max)} \end{aligned}$$

min: $(-1, -38)$
max: $(7, 218)$

10. The function $f(x) = 2mx^3 + 3nx^2 + qx + p$ has one point of inflection. Determine an expression for the slope of the tangent at that point of inflection. Assume m, n, q and p are all constants.

$$\begin{aligned} f'(x) &= 6mx^2 + 6nx + q \\ f''(x) &= 12mx + 6n \\ &= 6(2mx + n) \rightarrow x = \frac{-n}{2m} \end{aligned}$$

this is tangent

$$y = \left(\frac{-3n^2}{2m} + q \right) x + b$$

$$\begin{aligned} f'\left(\frac{-n}{2m}\right) &= 6m\left(\frac{-n}{2m}\right)^2 + 6n\left(\frac{-n}{2m}\right) + q + p \\ &= 6m\left(\frac{n^2}{4m^2}\right) - \frac{6n^2}{2m} + q + p \\ &= \frac{3n^2}{2m} - \frac{6n^2}{2m} + q \\ &= \frac{-3n^2}{2m} + q \end{aligned}$$

this is slope of the tangent